Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
MCA | Sem-4 | Winter-2023

Course Code	:	2105CS412	**Date**	:	16-10-2023
Course Name	:	Computer Networks	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Explain following terms: 1) Loss 2) Throughput **4**

(B) Give difference between LAN, MAN and WAN. **3**

OR

Explain any two network topologies with example.

(C) Explain following terms: 1) Propagation Delay 2) Transmission Delay **7**
And solve the following example. 3) How long does it take a packet of length 1,000 bytes to propagate over a link of distance 2,500 km, propagation speed 2.5 * 10^8 m/s, and transmission rate 2 Mbps?

OR

What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?

Q.2 **(A)** At which OSI layer are following devices used? **4**
1) Hub 2) Switch 3) Bridge 4) Gateway

(B) Explain co-axial cable in brief. **3**

OR

Explain twisted pair cable in brief.

(C) Discuss different types of guided and unguided media used to transmit data in network. **7**

OR

Explain the role of fiber optics in computer networks. How a fiber optic network is formed?

Q.3 **(A)** A bit stream 11010 is transmitted using standard CRC method. The generator polynomial is x^3+x+1. Calculate CRC code. **4**

(B) Explain Unicast, Multicast and Broadcast. **3**

OR

Differentiate between connection oriented service and connectionless service.

(C) Draw and explain Ethernet 802.3 Frame structure. **7**

OR

Explain CSMA/CD protocol in detail.

Q.4	**(A)**	If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts possible per subnet?	**4**
	(B)	How does DHCP protocol works? Explain.	**3**

OR

List the classes in classful addressing

	(C)	Explain IPV4 header in detail.	**7**

OR

Explain Distance Vector Routing protocol in detail.

Q.5	**(A)**	Why distributed database design is more preferred over centralized design to implement DNS in the Internet? Justify.	**4**
	(B)	What is HTTP? Explain its GET and POST methods.	**3**

OR

Give Difference between UDP and TCP.

	(C)	What do you mean by congestion? Explain the slow-start component of the TCP congestion-control algorithm.	**7**

OR

Describe Go Back N and Selective Repeat protocol.

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